

EXPERIMENT NO. 3

NEWTON'S LAW OF COOLING

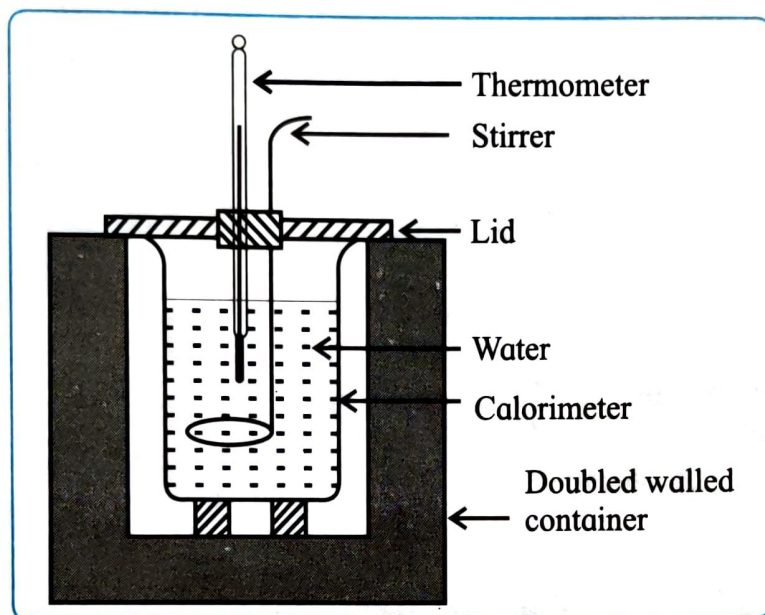
Aim : To study the relationship between the temperature of a hot body and time by plotting a cooling curve.

Apparatus : A thimble or calorimeter, constant temperature enclosure, a thermometer, stop clock, hot water bath etc.

Newton's law of cooling :

The rate of fall of temperature $\left(\frac{d\theta}{dt}\right)$ by a body is directly proportional to the difference of temperature $(\theta - \theta_0)$ of the body over the surroundings, provided the difference is small.

Diagram :



Theory:

The rate at which a hot body loses heat is directly proportional to the difference between the temperature of the hot body and that of its surroundings and depends on the nature of material and the surface area of the body. This is Newton's law of Cooling

Rate of loss of heat $\frac{d\theta}{dt} = K (\theta - \theta_0)$

Where, K = is the constant of proportionality $K = \frac{K'}{ms}$

also, $\frac{d\theta}{dt} = ms \frac{d\theta}{dt}$

$\therefore ms \frac{d\theta}{dt} = -K' (\theta - \theta_0)$

$\therefore \frac{d\theta}{dt} = -\frac{k'}{ms} (\theta - \theta_0)$

Formula :

$\frac{d\theta}{dt} = -K (\theta - \theta_0)$

$K' = k/ms$ is another constant it is water equivalent of the calorimeter negative fig indicates that loss of heat implies temperature decrease.

Procedure : experiment without wooden box.

1. Heat the water in water bath.
2. Note down the temperature of the surrounding (room temperature) by using thermometer.
3. Fill the thimble or calorimeter with hot water about 75 to 80°C upto $\frac{3}{4}$ th of its capacity.
Insert thermometer such that its bulb is well inside the hot water.
4. Note the initial temperature which is about 75° to 80°C above the room temperature.
5. start the stop watch.
6. Note the temperature after every minutes.
7. Take required number of readings till temperature of water falls about 40°C .
8. Plot the graph of temperature ($\theta_0^{\circ}\text{C}$) against time (t) in minutes. Draw smooth curve through the points.
9. Plot the graph of $\frac{d\theta}{dt}$ against temperature θ . Hence determine the room temperature (θ_0).
10. Plot a graph of rate of cooling against excess temperature above the surrounding Hence verify Newton's law of cooling.

Observations :

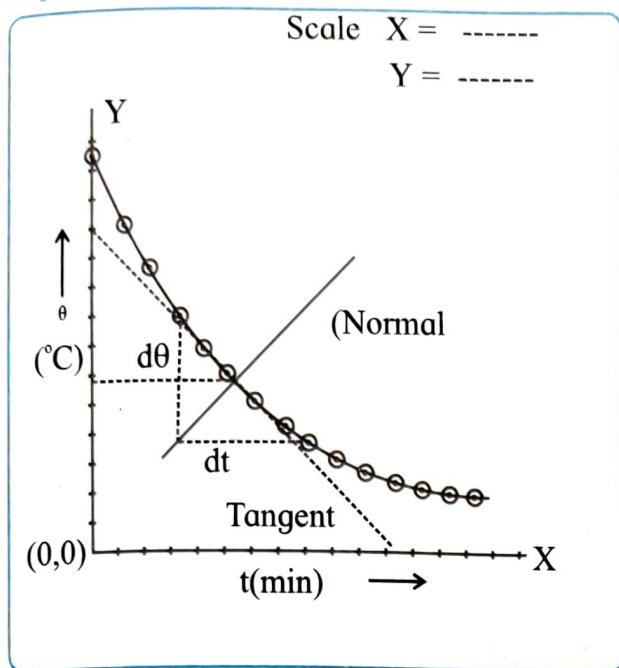
1. Least count of stop watch = $\frac{1}{1000}$ s.
2. Least count of thermometer = $\frac{1}{100}$ $^{\circ}\text{C}$.
3. Temperature of the surrounding (θ_0) = 30°C .

Observation Table (I) :

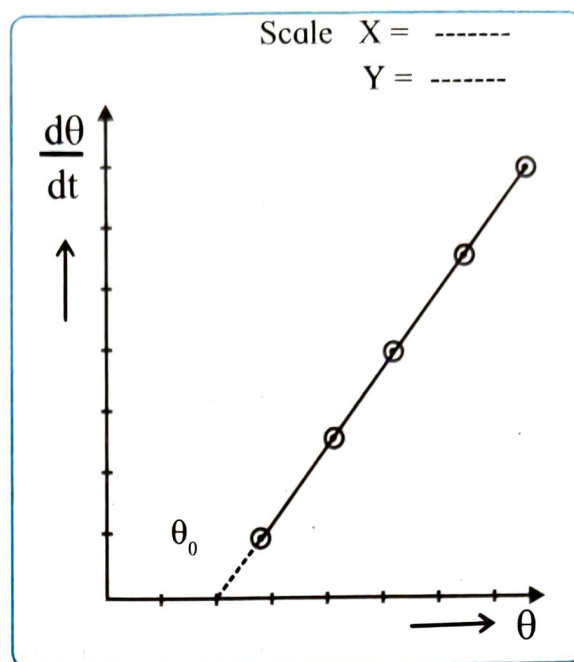
Ob. No.	Time in min	Temperature of water $^{\circ}\text{C}$	Ob. No.	Time in min	Temperature of water $^{\circ}\text{C}$
1	0	90	16	15	74
2	1	89	17	16	73
3	2	88	18	17	72
4	3	87	19	18	71
5	4	86	20	19	70
6	5	85	21	20	69
7	6	84	22	21	68
8	7	82	23	22	67
9	8	81	24	23	66
10	9	80	25	24	65
11	10	79	26	25	64
12	11	78	27	26	63
13	12	77	28	27	62
14	13	76	29	28	61
15	14	75	30	29	60

Graph of Temperature vs. Time.

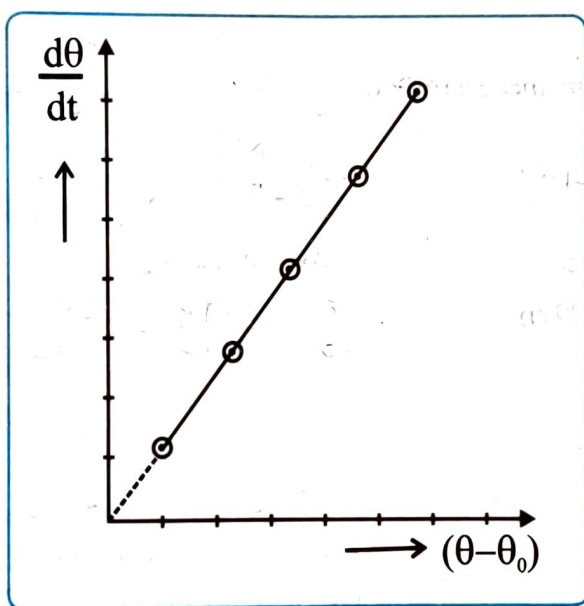
Graph I, II & III



Cooling curve



Graph of $(d\theta/dt)$ versus θ



Graph of $(d\theta/dt)$ versus $(\theta - \theta_0)$

Observation Table From cooling curve (II) : $\theta_0 = 32^{\circ}\text{C}$ (Room temp.)

Obs. No.	Temp $\theta^{\circ}\text{C}$	$\frac{d\theta}{dt}$ $^{\circ}\text{C}/\text{min}$	$(\theta - \theta_0)$
1	84.5	1.66	54.5
2	78	1.5	48
3	73	0.66	43
4	68	1	38
5	62	1	32

Result :

1. The nature of the graph of temperature against time is a curve. It is known as Newton's cooling curve.
2. This cooling curve will be steep at first, after it will become less steep as the temperature approaches to temperature of the surrounding.
3. As the graph of $\frac{d\theta}{dt}$ against $(\theta - \theta_0)$ is straight line, Passing through origin Newton's law of cooling is verified.
4. Room temperature by graph, $\theta_0 = \text{-----}^\circ\text{C}$

Precautions :

- i) Make sure that the bulb of thermometer is well inside the water.
- ii) The enclosure should have proper insulation to avoid loss of heat due to conduction or convection from hot water.
- iii) The water in the calorimeter should be gently stirred continually.

Questions

1. Certain quantity of water cools from 70°C to 60°C in the first 5 min and 54°C in the next 5 min. Then what will be the temperature of the surrounding (Ans. 45°C).

$$\begin{aligned} -K(65-T) \quad 60-70 &= -10 \\ -K(51-T) \quad \therefore 285-5T &= 195-3T \\ \therefore 90 &= 2T \\ \therefore T &= 45^\circ\text{C} \end{aligned}$$
$$\frac{-10}{-6} = \frac{65-T}{51-T}$$

2. You take an ice-cream out of the freezer, keep at -18°C . outside it is 32°C . After one minute, the ice cream has warmed to -8°C what is the temperature of the icecream after five minutes? (Ans. $\approx 15.6^\circ\text{C}$).

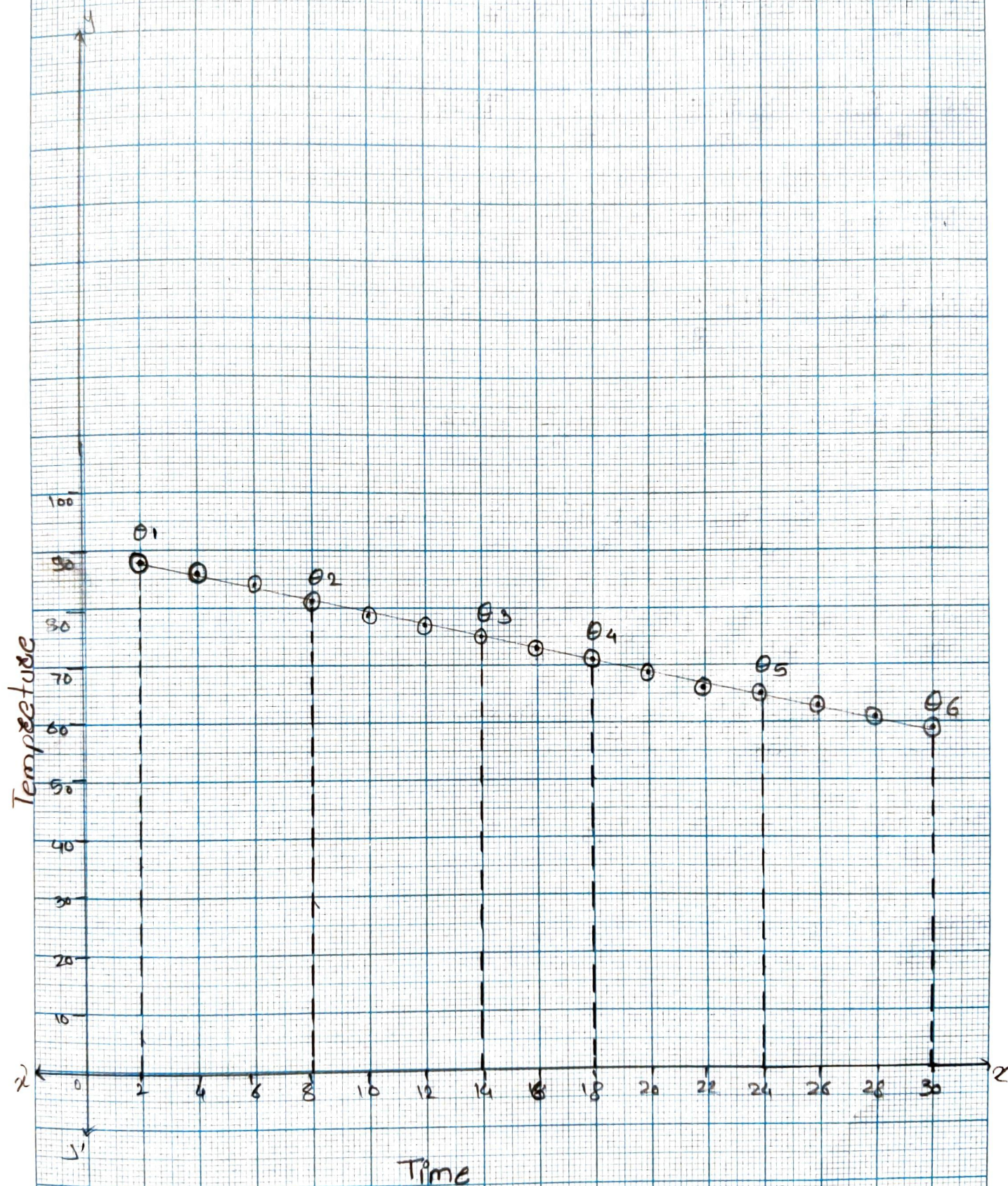
$$\begin{aligned} \text{for } t=1 \text{ min.} \quad \ln(e^{-K}) &= \ln(0.8) \quad T(5) = 32 + (-18-32) \times \\ -8 &= 32 + (-18-32) \times e^{-K} \quad -K = \ln(0.8) \quad e^{(0.2231 \times 5)} \\ e^{-K} &= (32+8)/50 \quad \text{multiply by } -1 \quad T(5) = 32 - 50 \times \\ \therefore e^{-K} &= 40/50 \quad \text{to find } K \quad 0.3274 \\ e^{-K} &= 0.8 \quad K = -\ln(0.8) \quad T(5) = 32 - 16.37 \\ &\approx 0.2231 \quad T(5) = 15.63^\circ\text{C} \end{aligned}$$

3. State Newton's law of cooling. write its expression in the form of specific heat.

The rate of fall of temperature ($d\theta/dt$) by body is directly proportional to the difference of temp. $(\theta - \theta_0)$ of the body over the surroundings, provided the difference is small. $d\theta/dt = -K(\theta - \theta_0)$

Remark and sign of teacher :

Scale: 1 cm = 25 sec (time)
 1 cm = 10% (temp)



Sign of teacher

$$1) \frac{\theta_1 + \theta_2}{2} = \frac{88 + 81}{2} = \frac{169}{2} = 84.2$$

$$2) \frac{\theta_2 + \theta_3}{2} = \frac{81 + 75}{2} = \frac{156}{2} = 78$$

$$3) \frac{\theta_3 + \theta_4}{2} = \frac{75 + 71}{2} = \frac{146}{2} = 73$$

$$4) \frac{\theta_4 + \theta_5}{2} = \frac{71 + 65}{2} = \frac{136}{2} = 68$$

$$5) \frac{\theta_5 + \theta_6}{2} = \frac{65 + 59}{2} = \frac{124}{2} = 62$$

$$6) \frac{d\theta_1}{dt} = \frac{\theta_1 - \theta_3}{dt} = \frac{81 - 75}{6} = \frac{6}{6} = 1.16$$

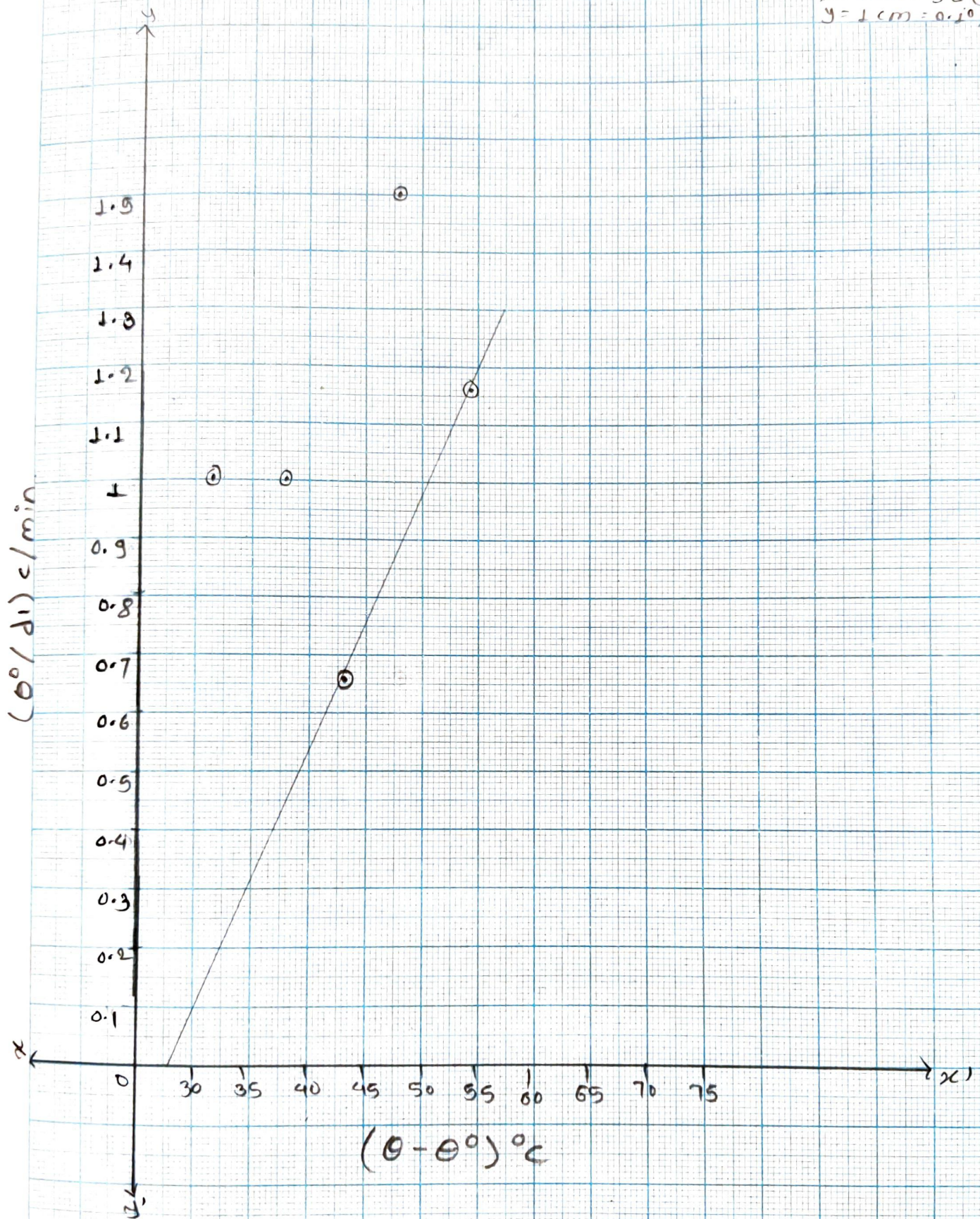
$$7) \frac{d\theta_2}{dt} = \frac{\theta_2 - \theta_3}{dt} = \frac{4}{6} = 0.66$$

$$8) \frac{d\theta_3}{dt} = \frac{\theta_3 - \theta_4}{dt} = \frac{4}{6} = 0.66$$

$$9) \frac{\theta_4 - \theta_5}{dt} = \frac{71 - 65}{6} = \frac{6}{6} = 1$$

$$10) \frac{\theta_5 - \theta_6}{dt} = \frac{65 - 59}{6} = \frac{6}{6} = 1$$

Scale :
 $x = 1 \text{ cm} = 5^\circ \text{C} (\theta - \theta^0)$
 $y = 1 \text{ cm} = 0.1^\circ / \text{min}$



Sign of teacher